

Four canonical monoidal structures on **Cont**

the classification behind the census, their closures, and how they interact

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Abstract

The category of containers carries four monoidal structures that everyone writes down: the coproduct $+$, the product $\times$, the Dirichlet tensor $\otimes$, and the sequential (substitution) operator $\triangleleft$. We give each its associator and unitors explicitly, in container language, and record that all four are machine-checked as monoidal categories in Lean 4. Then we ask the question the census leaves open — *why these four?* — and answer it. Every monoidal structure on **Set** induces one on **Cont** by Day convolution [Cited: Niu–Spivak]; $\times$ and $\otimes$ are the images of $(\mathbf{Set}, +, \emptyset)$ and $(\mathbf{Set}, \times, 1)$, while $+$ and $\triangleleft$ lie outside that family. Our first theorem says the Day construction is not merely a source of examples but an *equivalence* onto the monoidal structures that preserve coproducts in each variable and close up the representables. Our second says that the categorical product is the *unique* monoidal structure on **Cont** — among all of them, Day or not — whose extension is the pointwise product of functors; being pointwise is the rare thing, not the generic one. Our third says what the comparison map $\otimes \rightarrow \triangleleft$ is: the counit of a coreflection, exhibiting $p \otimes (-)$ as the terminal coproduct-preserving approximation to $p \triangleleft (-)$. The Dirichlet tensor is the sequential operator with its dependency switched off, and not by fiat. We then close the structures — the internal hom for $\otimes$ [Cited: Niu–Spivak] is machine-checked in Lean 4, we believe the first such certification — and study how they interact through the normal duoidal category $(\mathbf{Cont}, \otimes, \triangleleft)$, whose spine is the comparitor. There the double comonoids are classified: they are exactly the *sets of commutative monoids*, by a fibrewise Eckmann–Hilton collapse, correcting a plausible guess in both directions, on which the chapter closes. (A further step outside the Day family altogether — identifying two uninterpreted tensors of Dorta–Jarvis–Niu as the multiplicative and directed Dialectica products of Gödel’s functional interpretation — is deferred to a later **Cont**(C) chapter and held in a companion file.)

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1 Introduction

The census, and the question it leaves open

Open any account of containers, or of the polynomial functors they present, and you meet four monoidal structures on the same category:

	shapes of $p \odot q$	positions there	unit
coproduct $+$	$S_p + S_q$	inherited	0
product $\times$	$S_p \times S_q$	$p[s] + q[t]$	$1 = y^\emptyset$
Dirichlet $\otimes$	$S_p \times S_q$	$p[s] \times q[t]$	$y = y^1$
sequential $\triangleleft$	$\sum_{s \in S_p} (p[s] \rightarrow S_q)$	$\sum_{i \in p[s]} q[f\ i]$	y

This is a *census*. It tells you what is there; it does not tell you why, and it does not tell you how the four relate. A reader who has just been handed this table is entitled to three complaints, and each of them is a mathematical question.

First: the product's positions are a *disjoint union* — yet its extension is the *pointwise product* of functors. The Dirichlet tensor multiplies the positions — and its extension is not pointwise at all. Why is the arithmetic upside down?

Second: why exactly four? Is four a fact about **Cont**, or an accident of what people happen to have written down?

Third: $\otimes$ and $\triangleleft$ have the same unit y , and on one-shape containers they visibly agree: $y^A \otimes y^B = y^{A \times B} = y^A \triangleleft y^B$. Two different tensors that coincide on representables — is that a coincidence?

None of the three has a satisfying answer at the level of the census. All three have the same answer one level up.

The three theorems

Write y^A for the container with one shape and position set A (the *representable*, extension $X \mapsto X^A$), and call a monoidal structure $(\odot, J)$ on **Cont** **convolutional** when

- (D1) $\odot$ preserves coproducts in each variable separately (including the empty one: $p \odot 0 \cong 0$), and
- (D2) the representables are closed under $\odot$.

- **Theorem A (classification, §4).** The Day construction $(\star, I) \mapsto (\odot_\star, y^I)$ is an *equivalence* from monoidal structures on **Set** to convolutional monoidal structures on **Cont**. Every convolutional structure is a Day convolution, and the structure on **Set** inducing it is unique up to monoidal isomorphism.
- **Theorem B⁺ (uniqueness of the pointwise structure, §5).** Let $(\odot, J)$ be *any* monoidal structure on **Cont** — convolutional or not. If $\llbracket p \odot q \rrbracket \cong \llbracket p \rrbracket \times \llbracket q \rrbracket$ naturally, then $(\odot, J) \cong (\times, 1)$. The categorical product is the *only* monoidal structure on containers whose extension is the pointwise product.
- **Theorem C (what the comparitor is, §7).** For each p , the functor $p \otimes (-)$ is the left Kan extension of $p \triangleleft (-)$ along the representable embedding $y^{(-)}: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Cont}$, and the known comparison map $o_{p,q}: p \otimes q \rightarrow p \triangleleft q$ is the *counit of the resulting coreflection*. Hence $p \otimes (-)$ is the terminal coproduct-preserving approximation to $p \triangleleft (-)$.

These answer the three complaints in order. The arithmetic is upside down because $\times$ is the Day convolution of the *coproduct* of sets, and $y^{A+B} \cong y^A \times y^B$ is the exponential law (§3). Four is an accident: **Cont** carries $+$, it carries $\triangleleft$, and it carries a Day family indexed by the monoidal structures on **Set** — a family that is a proper class (§6). And $\otimes$ agrees with $\triangleleft$ on representables because it is *generated* by that agreement: $\otimes$ is the Day-ification of $\triangleleft$ (§7).

Two further movements: closure, and interaction

The census answered, two questions a monoidal category always raises remain. *Are the tensors closed? And how do they interact with one another?* Both have clean answers, and both are largely a matter of *organising* existing results correctly rather than proving new ones — so we are scrupulous in §§8–9 about what is cited and what is ours. (A further movement, leaving the Day family behind entirely to locate its outer boundary via the Dialectica tensors, is deferred to a later **Cont(C)** chapter; the material is held in `papers/ductectica-tensors-deferred.tex`.)

Closure (§8). The Dirichlet tensor is closed: its internal hom is the container whose *shapes are the container morphisms*, and this is in the literature [Cited: Niu–Spivak, Spivak]. Our contribution here is not the formula but its *certification*: $(\mathbf{Cont}, \otimes, y)$ is a closed monoidal category in Lean 4, currying and uncurrying inverse on the nose, every proof `rfl`. To our knowledge this is the first machine-checked Dirichlet closure. The sequential operator $\triangleleft$ is not closed on the left, but has a right coclosure [Cited: Spivak; Meyers], and **Cont** is cartesian closed but *not* locally cartesian closed [Cited: Altenkirch–Levy–Staton] — so it does not model dependent type theory, a fact worth stating plainly.

Interaction (§9). The pair $(\otimes, \triangleleft)$ is a *normal duoidal* category [Cited: Spivak–Srinivasan], and its interchange law is the home of the comparitor. Duoidal categories have *double comonoids*, objects that are comonoids for both tensors compatibly; here the $\triangleleft$ -comonoids are the small categories, so a double comonoid is a small category with extra structure. We classify them: **the double comonoids in $(\mathbf{Cont}, \triangleleft, \otimes)$ are exactly the sets of commutative monoids** (Theorem 9.2). The mechanism is a fibrewise Eckmann–Hilton collapse, and the classification corrects a natural guess — “the degenerate polynomials” — in both directions.

A correction, stated plainly

An earlier draft of this material (June 2026) contained two claims that are false, and since they were circulated I state their refutations here rather than quietly deleting them.

1. “ $\llbracket - \rrbracket$ is strict monoidal for $\otimes$.” It is **strong**, not strict. The argument was circular: it defined the Dirichlet convolution on functors by *reading off the container presentation* — that is, by transporting the tensor along $\llbracket - \rrbracket$ — and then observed that $\llbracket - \rrbracket$ preserved it. Intrinsically $\otimes$ is a Day convolution, defined by a coend, and the comparison is the co-Yoneda isomorphism, not an identity. Corollary 5.7 below explains why no repair was possible: $\llbracket - \rrbracket$ is monoidal into the pointwise product *only* for the cartesian structure.
2. “ $\otimes$ fails to be pointwise because it is a Day convolution.” False, and the reason is instructive: $\times$ is *also* a Day convolution — of $(\mathbf{Set}, +, \emptyset)$ — and it *is* pointwise. Day-ness is not the criterion, and there is *no* “pointwise/closed duality” of the kind that draft proposed. The criterion is whether the corepresentable splits the tensor of $\mathbf{Set}$: $y^{A+B} \cong y^A \times y^B$ does split, $y^{A \times B}$ does not. Witness: $y^3 \otimes y^3 = y^9$, against a pointwise y^6 . (This splitting criterion — which *survives* — reads almost identically to the false duality, and one must not confuse them: the survivor is about the corepresentable splitting the *domain* tensor, not about the internal hom preserving anything.)
3. “The pentagon for $\triangleleft$ is trivial.” A bad word for a true fact. The coherence of $\triangleleft$ discharges by `rfl` in Lean because both bracketings of an iterated $\triangleleft$ flatten to the *same* Σ -telescope (§3.3); that is a *syntactic* coincidence of normal forms, not a claim that parenthesisation is vacuous. Mac Lane’s theorem is not *free* for $\triangleleft$ — it is *pre-paid*: the work is done by the definitional equality, once, rather than by a coherence argument each time. We restate the fact carefully in §3.3 rather than call it trivial.

The identification “Dirichlet tensor = Day convolution” is *not ours*; it is Niu–Spivak [3, Prop. 3.79]. What is new here is the converse — Theorem A — and what the converse buys: Theorems B^+ and C.

Plan

Everything rests on one structural fact, proved in §2: **Cont** is the free coproduct completion of $\mathbf{Set}^{\text{op}}$, so a coproduct-preserving functor out of **Cont** — and any natural transformation between two such — is determined by what it does to representables. Given that, the three theorems are elementary. Section 3 is the census itself; §§4–7 prove A, B^+ and C, with §6 showing that no cheaper invariant than pointwiseness could have isolated the product. Section 8 closes the tensors; §9 studies their interaction and classifies the double comonoids, on which this chapter closes. (The step outside the Day family altogether — identifying two further tensors of Dorta–Jarvis–Niu as the Dialectica products of Gödel’s functional interpretation — is deferred to a later **Cont**(C) chapter, held in `papers/ductectica-tensors-deferred.tex`.) Section 10 records provenance, what is machine-checked, and what is open.

Every statement carries a bracketed provenance tag, following the book’s convention: [Cited], [MacBeth], [MacBeth, Lean-verified: *file*], [Open].

2 Preliminaries

Definition 2.1 (Container, extension). A *container* is a pair $p = (S_p, p[-])$ with S_p a set of *shapes* and $p[s]$ a set of *positions* for each $s \in S_p$. A morphism $p \rightarrow q$ is a pair (f, g) with

$f: S_p \rightarrow S_q$ on shapes and, contravariantly, $g_s: q[f\ s] \rightarrow p[s]$ on positions. The *extension* is

$$\llbracket p \rrbracket X = \sum_{s \in S_p} X^{p[s]},$$

and $\llbracket - \rrbracket: \mathbf{Cont} \rightarrow [\mathbf{Set}, \mathbf{Set}]$ is fully faithful, with image the polynomial functors **Poly**. [Cited: Abbott–Altenkirch–Ghani 2005]

We write $p = \sum_{s \in S_p} y^{p[s]}$ and use **Cont** and **Poly** interchangeably. Two families of containers do all the work below.

Definition 2.2 (Representable, linear). y^A is the container with a single shape and positions A ; so $\llbracket y^A \rrbracket X = X^A$. A container is *representable* if it has exactly one shape. We abbreviate $y := y^1$ (extension the identity functor), $1 := y^\emptyset$ (extension the constant functor at a point — the terminal container), and write 0 for the container with no shapes. A container is *linear* if every position set $p[s]$ is a singleton.

Lemma 2.3 (The representable embedding is fully faithful). $y^{(-)}: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Cont}$, $A \mapsto y^A$, is fully faithful. [MacBeth (elementary)]

Proof. A morphism $y^A \rightarrow y^B$ is a map on shapes $1 \rightarrow 1$, which is forced, together with a map $B \rightarrow A$ on positions. So $\mathbf{Cont}(y^A, y^B) \cong \mathbf{Set}(B, A) = \mathbf{Set}^{\text{op}}(A, B)$, naturally in A and B . $\square$

Lemma 2.4 (**Cont** is the free coproduct completion of $\mathbf{Set}^{\text{op}}$). $\mathbf{Cont} \cong \mathbf{Fam}(\mathbf{Set}^{\text{op}})$, an isomorphism of categories, with $y^{(-)}$ as the universal embedding. [Folklore; proof MacBeth]¹

Proof. An object of $\mathbf{Fam}(\mathcal{C})$ is a family $(A_i)_{i \in I}$ of objects of $\mathcal{C}$; a morphism $(A_i)_I \rightarrow (B_j)_J$ is a function $u: I \rightarrow J$ together with morphisms $A_i \rightarrow B_{u\ i}$ in $\mathcal{C}$. Take $\mathcal{C} = \mathbf{Set}^{\text{op}}$. An object is then a set I together with an I -indexed family of sets — exactly a container. A morphism is a function $u: I \rightarrow J$ together with $\mathbf{Set}^{\text{op}}$ -maps $A_i \rightarrow B_{u\ i}$, that is, $\mathbf{Set}$ -maps $B_{u\ i} \rightarrow A_i$ — exactly a container morphism (Definition 2.1), variance included. $\square$

Lemma 2.5 (Universal property). Let $\mathcal{E}$ have small coproducts. Restriction along $y^{(-)}$ is an equivalence

$$\text{Fun}_{\sqcup}(\mathbf{Cont}, \mathcal{E}) \simeq \text{Fun}(\mathbf{Set}^{\text{op}}, \mathcal{E}),$$

where $\text{Fun}_{\sqcup}$ denotes coproduct-preserving functors; the inverse sends G to $\widehat{G}(p) = \sum_{s \in S_p} G(p[s])$. More generally, for each $n \geq 1$, restriction is an equivalence

$$\text{Fun}_{\sqcup, \dots, \sqcup}(\mathbf{Cont}^n, \mathcal{E}) \simeq \text{Fun}((\mathbf{Set}^{\text{op}})^n, \mathcal{E})$$

between functors preserving coproducts separately in each variable and arbitrary n -ary functors on $\mathbf{Set}^{\text{op}}$. [Folklore; proof MacBeth]

Proof. The case $n = 1$ is the universal property of **Fam** (Lemma 2.4). For $n = 2$: coproducts in $\text{Fun}_{\sqcup}(\mathbf{Cont}, \mathcal{E})$ are computed pointwise, so by currying, a bifunctor $\mathbf{Cont} \times \mathbf{Cont} \rightarrow \mathcal{E}$ preserving coproducts in the second variable is a functor $\mathbf{Cont} \rightarrow \text{Fun}_{\sqcup}(\mathbf{Cont}, \mathcal{E})$, and it preserves coproducts in the first variable exactly when that functor does. Hence

$$\begin{aligned} \text{Fun}_{\sqcup, \sqcup}(\mathbf{Cont}^2, \mathcal{E}) &\simeq \text{Fun}_{\sqcup}(\mathbf{Cont}, \text{Fun}_{\sqcup}(\mathbf{Cont}, \mathcal{E})) \simeq \text{Fun}(\mathbf{Set}^{\text{op}}, \text{Fun}_{\sqcup}(\mathbf{Cont}, \mathcal{E})) \\ &\simeq \text{Fun}((\mathbf{Set}^{\text{op}})^2, \mathcal{E}), \end{aligned}$$

each step by the case $n = 1$. Induct. $\square$

¹The identification of **Poly** with the free coproduct completion of $\mathbf{Set}^{\text{op}}$ is standard in the polynomial-functor literature and we claim no originality for it. We give the elementary proof because the universal property (Lemma 2.5) is used repeatedly and we want it in the exact form we use. Precise locators are deferred to a citation pass; the reader should treat this as folklore, not as a MacBeth theorem.

We use Lemma 2.5 constantly and in one specific form, worth stating as a slogan:

A functor $\mathbf{Cont}^n \rightarrow \mathbf{Cont}$ that preserves coproducts in each variable is determined, up to canonical isomorphism, by its restriction to representables — and so are the natural transformations between two such. An equation between such transformations (a pentagon, a triangle) holds if and only if it holds on representables.

Lemma 2.5 compares functors that already preserve coproducts. In §7 we shall need to compare a coproduct-preserving functor with one that does *not* preserve coproducts (namely $p \triangleleft -$), so we record the Kan extension itself, and not merely the equivalence.

Lemma 2.6 (Extension by coproducts is the left Kan extension). *Write $J := y^{(-)}$. For every $G: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Cont}$ the left Kan extension of G along J exists and is given by*

$$(\text{Lan}_J G)(q) = \sum_{t \in S_q} G(q[t]).$$

Consequently $\text{Lan}_J \dashv \text{res}_J$, where $\text{res}_J F = F \circ J$ — and this adjunction holds against arbitrary $F: \mathbf{Cont} \rightarrow \mathbf{Cont}$, not merely coproduct-preserving ones. [Folklore; proof MacBeth]

Proof. The pointwise formula computes $(\text{Lan}_J G)(q)$ as the colimit of G over the comma category J/q . An object of J/q is a set A together with a morphism $y^A \rightarrow q$, that is (Definition 2.1) a shape $t \in S_q$ and a function $u: q[t] \rightarrow A$. A morphism $(A, t, u) \rightarrow (A', t', u')$ is a morphism $A \rightarrow A'$ of $\mathbf{Set}^{\text{op}}$, i.e. a function $g: A' \rightarrow A$; the commuting condition forces $t = t'$ (the shape map of $y^A \rightarrow y^{A'}$ is the only one there is, so the composite still selects t') and $u = g \circ u'$.

So J/q is the disjoint union over $t \in S_q$ of a fibre $\mathcal{F}_t$ whose objects are functions $u: q[t] \rightarrow A$ and whose morphisms $(A, u) \rightarrow (A', u')$ are $g: A' \rightarrow A$ with $g \circ u' = u$. That is, $\mathcal{F}_t = (q[t]/\mathbf{Set})^{\text{op}}$, and the pair $(q[t], \text{id})$ — initial in the coslice — is *terminal* in $\mathcal{F}_t$. A colimit over a category with a terminal object is the value of the diagram there, so $\text{colim}_{\mathcal{F}_t} G = G(q[t])$ and $\text{colim}_{J/q} G = \sum_{t \in S_q} G(q[t])$. This exists for every G and every q , so the pointwise left Kan extension exists, and a pointwise left Kan extension along J that exists at every object is left adjoint to restriction along J . $\square$

Remark 2.7 (No size problem). The usual definition of Day convolution on $\mathbf{Poly}$ is a coend over $\mathbf{Set}^2$, which is large [3, eq. (3.80)]. We never form a coend. Lemma 2.5 is a *freeness* statement, and the extension it produces is a coproduct indexed by $S_p \times S_q$, which is small. Nothing below needs the coend to exist; co-Yoneda [3, eq. (3.81)] says the two agree when it does.

3 The census: four structures, and why the arithmetic looks upside down

Definition 3.1 (The four). For containers p, q :

$$\begin{aligned} p + q &:= (S_p + S_q, \text{ positions inherited}), & \text{unit } 0; \\ p \times q &:= (S_p \times S_q, (s, t) \mapsto p[s] + q[t]), & \text{unit } 1 = y^\emptyset; \\ p \otimes q &:= (S_p \times S_q, (s, t) \mapsto p[s] \times q[t]), & \text{unit } y = y^1; \\ p \triangleleft q &:= \left(\sum_{s \in S_p} (p[s] \rightarrow S_q), (s, f) \mapsto \sum_{i \in p[s]} q[f\,i] \right), & \text{unit } y. \end{aligned}$$

On extensions, $\llbracket p + q \rrbracket = \llbracket p \rrbracket + \llbracket q \rrbracket$ and $\llbracket p \times q \rrbracket = \llbracket p \rrbracket \times \llbracket q \rrbracket$ (both pointwise) and $\llbracket p \triangleleft q \rrbracket = \llbracket p \rrbracket \circ \llbracket q \rrbracket$. There is *no* pointwise description of $\llbracket p \otimes q \rrbracket$. [Cited: Spivak 2022 [4]; Niu–Spivak [3]]

That last sentence is the seed of everything that follows. But first, the complaint a new reader always makes.

3.1 The first jewel: a sum of positions gives a product of functors

The product of containers has *shapes* $S_p \times S_q$ — fine, choose one of each — but its *positions* are $p[s] + q[t]$, a *disjoint union*. Readers find this perverse: surely a product of things should multiply something. It does not, and the reason is one line of arithmetic:

$$\llbracket p \times q \rrbracket X = \sum_{(s,t)} X^{p[s]+q[t]} = \sum_{(s,t)} X^{p[s]} \times X^{q[t]} = \left(\sum_s X^{p[s]} \right) \times \left(\sum_t X^{q[t]} \right) = \llbracket p \rrbracket X \times \llbracket q \rrbracket X.$$

The middle step is the *exponential law* $X^{A+B} \cong X^A \times X^B$: a labelling of the disjoint union of the two position sets is precisely a pair of labellings. To hold a p -structure and a q -structure side by side is to fill in p ’s positions *and* q ’s positions, and the positions you must fill are the two sets together. The sum on positions is not a quirk of the definition; it is what “both at once” means when the data live at positions.

Read it the other way and it is a warning. The Dirichlet tensor multiplies the positions, so

$$\llbracket p \otimes q \rrbracket X = \sum_{(s,t)} X^{p[s] \times q[t]},$$

and $X^{A \times B}$ does *not* split as anything built from X^A and X^B . So $\otimes$ cannot be pointwise — $y^3 \otimes y^3 = y^9$, where a pointwise product would give y^6 . Two tensors, the same shape set, opposite behaviour on extensions, and the whole difference is which operation on *positions* appears in the exponent. Section 4 explains why the exponent is exactly where the choice lives.

3.2 Coherence, and the second jewel

Theorem 3.2 (The four are monoidal categories). *Each of $(\mathbf{Cont}, +, 0)$, $(\mathbf{Cont}, \times, 1)$, $(\mathbf{Cont}, \otimes, y)$ and $(\mathbf{Cont}, \triangleleft, y)$ is a monoidal category. The first three are symmetric; $\triangleleft$ is not. [Cited: Spivak 2022 [4]; MacBeth, Lean-verified: `FourMonoidal.lean`, `Dirichlet.lean`, `Monoidal.lean`]*

The Lean development gives one `MonoidalCategory Container` instance per structure — `contCoproductMonoidal`, `contProductMonoidal`, `contDirichletMonoidal` and `ContMonoidal` — each with its associator, unitors, naturality, pentagon and triangle discharged.

For $+$ and $\times$ coherence is free: they are the categorical coproduct and product, and any category with finite coproducts and products is monoidal for them. For $\times$ and $\otimes$ there is a second and more informative derivation: both are Day convolutions, so their associator and unitors act *separately* on the shape sets and on the position sets, inherited from **Set**’s own $+$ and $\times$, and the pentagon and triangle are then **Set**’s pentagon and triangle applied twice over. Proposition 4.2 says this in the right generality. The sequential operator is the one that needs work, and we do it in §3.3.

Before that, a reversal worth recording, because it is the opposite of what we predicted.

Remark 3.3 (The second jewel: $\otimes$ is the *tamest* of the four). We spent a month calling the Dirichlet tensor “the subtle one”. In Lean it is the only one of the four that costs nothing: in `contDirichletMonoidal` every coherence obligation — associator naturality, both unitors, pentagon, triangle — closes by `rfl`. Its associator is the definitional η -law for pairs. The other three all need `Quot.sound` somewhere, because their position sets are built by *case analysis* (a coproduct, or a dependent sum) rather than by pairing, and case analysis has to be proved to respect equality where pairing does not.

So $\otimes$ is subtle *semantically* — it is the one whose extension is not pointwise — and utterly tame *syntactically*. The two kinds of difficulty are unrelated, and conflating them cost us a month and one false theorem (§1).

3.3 The sequential operator, explicitly

$\triangleleft$ is the operator this book cares about most — its comonoids are the directed containers, hence the small categories (a theorem of Ahman–Uustalu [9]; in $\triangleleft$ -comonoid form it is the $\mathcal{C} = 1$ case of Dorta–Jarvis–Niu [10, Thm. 4.3]) — so we give its coherence data in full, in container language, rather than citing it.

Proposition 3.4 (Associator and unitors for $\triangleleft$). $\alpha: (p \triangleleft q) \triangleleft r \xrightarrow{\sim} p \triangleleft (q \triangleleft r)$ is given on shapes by currying and on positions by Σ -reassociation. In detail: a shape of $(p \triangleleft q) \triangleleft r$ is $((s, f), g)$ with $f: p[s] \rightarrow S_q$ and $g: \sum_{i \in p[s]} q[f\ i] \rightarrow S_r$; a shape of $p \triangleleft (q \triangleleft r)$ is (s, k) with $k: p[s] \rightarrow S_{q \triangleleft r}$; and

$$\alpha_s((s, f), g) = (s, k), \quad k\ i = (f\ i, j \mapsto g(i, j)),$$

while both sides have positions $\sum_i \sum_{j \in q[f\ i]} r[g(i, j)]$, bracketed $((i, j), m)$ on the left and $(i, (j, m))$ on the right, and $\alpha_p(i, (j, m)) = ((i, j), m)$. The unitors are

$$\lambda_s(*, g) = g\bullet, \quad \lambda_p(m) = (\bullet, m); \quad \rho_s(s, !) = s, \quad \rho_p(i) = (i, \bullet).$$

[MacBeth, Lean-verified: *Monoidal.lean*]

Proof. α_s is the composite of the distributivity equivalence $\prod_i \sum_t (q[t] \rightarrow S_r) \cong \sum_f \prod_i (q[f\ i] \rightarrow S_r)$ with currying $\prod_i (q[f\ i] \rightarrow S_r) \cong (\sum_i q[f\ i] \rightarrow S_r)$; its inverse sends (s, k) to $((s, f), g)$ with $f\ i = \pi_1(k\ i)$ and $g(i, j) = \pi_2(k\ i)\ j$. Once $k\ i = (f\ i, j \mapsto g(i, j))$ is substituted, the two position sets are the same iterated sum with different bracketing, and α_p is the canonical reassociation. Both components are bijections. The unitors are read off from $\sum_{i \in 1} q[g\bullet] \cong q[g\bullet]$ and $\sum_{i \in p[s]} 1 \cong p[s]$. $\square$

Both components are *transport-free*: in Lean the associator needs no **cast**, because currying and Σ -reassociation are definitional equalities of the underlying types. That is why the pentagon goes through by computation.

Theorem 3.5 (Pentagon and triangle for $\triangleleft$). *The associator and unitors of Proposition 3.4 satisfy the pentagon and the triangle identity. Hence $(\mathbf{Cont}, \triangleleft, y)$ is a monoidal category, and by Mac Lane’s coherence theorem [2] every formal diagram of associators and unitors commutes.* [MacBeth, Lean-verified: *Monoidal.lean* (*Container.pentagon*, *Container.triangle*)]

Proof. *Shapes:* both legs are a flattening. A shape of the source $((p_1 \triangleleft p_2) \triangleleft p_3) \triangleleft p_4$ is, unfolding Definition 3.1 three times, a tuple $W = (((s, f), g), h)$ with

$$f: p_1[s] \rightarrow S_2, \quad g: \{(i, j)\} \rightarrow S_3, \quad h: \{((i, j), m)\} \rightarrow S_4,$$

where $\{(i, j)\} = \sum_i p_2[f\ i]$ and $\{((i, j), m)\} = \sum_{(i, j)} p_3[g(i, j)]$. Define the *normal form*

$$\Phi(W) = (s, i \mapsto (f\ i, j \mapsto (g(i, j), m \mapsto h((i, j), m))))),$$

a shape of the target $p_1 \triangleleft (p_2 \triangleleft (p_3 \triangleleft p_4))$: it simply erases the bracketing and leaves the nested-function presentation. Computing the two legs of the pentagon with Proposition 3.4 — one leg applies $\alpha \triangleleft p_4$, then α , then $p_1 \triangleleft \alpha$; the other applies α twice — each yields $\Phi(W)$ after currying is unwound. Two maps equal to the same normal form are equal.

Positions: everything is the identity on the flat tuple. Every map in either leg is, on positions, a bijection that leaves the underlying flat tuple (i, j, m, n) alone and only changes its bracketing: α_p rebrackets one adjacent pair, and whiskering fixes the whiskered coordinate and transports the rest. Let fl denote the “erase brackets” bijection from either bracketed position set to the flat set $\{(i, j, m, n)\}$. Each elementary map μ satisfies $\text{fl} \circ \mu_p = \text{fl}$, hence so does any composite; so both legs equal $\text{fl}_{\text{tgt}}^{-1} \circ \text{fl}_{\text{src}}$, and they coincide.

The triangle is the same computation with $p_2 = y$: both sides send the shape $((s, f), g)$ of $(p_1 \triangleleft y) \triangleleft p_2$ to $(s, i \mapsto g(i, \bullet))$ and both send the position (i, m) to $((i, \bullet), m)$. $\square$

Remark 3.6. The proof of Theorem 3.5 is the informal shadow of what Lean does. In `Monoidal.lean` the pentagon and triangle close by `rfl` on both components: the normal form Φ is Lean’s normal form, and “erase the brackets” is definitional unfolding. Machine and hand agree on the mechanism, not merely on the conclusion.

4 Why these? The Day family and its classification

The census gives four. The next definition gives a proper class, and the four sort themselves inside it.

Definition 4.1 (The Day tensor of a monoidal structure on **Set**). Let $(\star, I)$ be a monoidal structure on **Set**. Define, on containers,

$$p \odot_{\star} q := \left(S_p \times S_q, (s, t) \mapsto p[s] \star q[t] \right) = \sum_{(s, t)} y^{p[s] \star q[t]}, \quad J_{\star} := y^I.$$

[Cited: Niu–Spivak [3, Prop. 3.79]]

Proposition 4.2 (Day tensors are monoidal). $(\odot_{\star}, J_{\star})$ is a monoidal structure on **Cont**; it preserves coproducts in each variable; and it sends representables to representables, with $y^A \odot_{\star} y^B = y^{A \star B}$. It is the Day convolution of $(\star, I)$. [Cited: Niu–Spivak [3, Prop. 3.79, eqs. (3.80)–(3.81)]]

Proof. The coherence data are inherited componentwise. The associator of $\odot_{\star}$ at (p, q, r) is the identity on shapes (both sides have shapes $S_p \times S_q \times S_r$) and, on positions, the associator of $\star$ at $(p[s], q[t], r[u])$ — reversed, positions being contravariant. Pentagon and triangle hold shapewise because they hold in **Set**. Coproduct preservation in each variable holds because $S_p \times (-)$ preserves coproducts of sets, including the empty one. Representables: immediate from the formula. The identification with Day convolution is [3, Prop. 3.79]. $\square$

Example 4.3 (Two of the four are Day tensors). $\odot_{(+, \emptyset)} = \times$ and $\odot_{(\times, 1)} = \otimes$. [Cited: Niu–Spivak [3, Prop. 3.79, ¶2]]

This is the answer to the upside-down arithmetic of §3: the positions of the categorical product are a disjoint union *because the product is the Day convolution of the coproduct of sets*. The tensor of **Set** lands in the exponent, and $y^{A+B} \cong y^A \times y^B$.

Lemma 4.4 (Every Day tensor annihilates 0). $p \odot_{\star} 0 \cong 0 \cong 0 \odot_{\star} p$. [MacBeth (elementary)]

Proof. $S_p \times \emptyset = \emptyset$. $\square$

Trivial as it is, Lemma 4.4 already expels the coproduct from the family: $p + 0 \cong p \not\cong 0$. So $+$ is not a Day tensor, and neither is $\triangleleft$ (Table 1). We now show that these two failures are the *only* way to fall outside the family.

4.1 Theorem A

Definition 4.5 (Convolutional). A monoidal structure $(\odot, J)$ on **Cont** is *convolutional* if (D1) $\odot$ preserves coproducts in each variable, including the empty coproduct ($p \odot 0 \cong 0$); and (D2) the representables are closed under $\odot$. [MacBeth]

Lemma 4.6 (The unit comes for free). (D1) and (D2) force J to be representable. [MacBeth]

Proof. Write $J = \sum_{t \in S_J} y^{J[t]}$. By (D1),

$$y^A \cong J \odot y^A = \sum_{t \in S_J} (y^{J[t]} \odot y^A),$$

and by (D2) each summand has exactly one shape. So the right-hand side has $|S_J|$ shapes and the left has one; hence $|S_J| = 1$. $\square$

So “convolutional” is a condition on the *tensor* alone — the unit has no say in the matter. This is what makes Theorem A a statement about bifunctors, and it is what lets Theorem B⁺ drop the Day hypothesis altogether.

Theorem 4.7 (A: classification). *The assignment $(\star, I) \mapsto (\odot_\star, y^I)$ is an **equivalence***

$$\{\text{monoidal structures on } \mathbf{Set}\} \simeq \{\text{convolutional monoidal structures on } \mathbf{Cont}\},$$

*both sides taken with monoidal isomorphisms over the identity functor as morphisms. In particular every convolutional monoidal structure on **Cont** is a Day convolution, and the monoidal structure on **Set** inducing it is unique up to monoidal isomorphism.* [MacBeth]

Proof. Well defined: Proposition 4.2.

Essentially surjective. Let $(\odot, J)$ be convolutional. By (D2) the full subcategory of representables is closed under $\odot$, and by Lemma 4.6 it contains J ; by Lemma 2.3 that subcategory is $\mathbf{Set}^{\text{op}}$. We need a canonical choice of tensor on **Set**, not merely “ $y^A \odot y^B$ is isomorphic to some y^C ”. By (D2) the container $y^A \odot y^B$ has exactly one shape, so we may put

$$A \star B := (y^A \odot y^B)[*],$$

the position set at that unique shape — and then $y^A \odot y^B = y^{A \star B}$ on the nose. Likewise $I := J[*]$, so $J = y^I$. Functoriality of $\star$ follows from that of $\odot$ together with Lemma 2.3. Applying (D2) twice, both $(y^A \odot y^B) \odot y^C$ and $y^A \odot (y^B \odot y^C)$ are representable, so the associator of $\odot$ at representables is an isomorphism *between representables* and hence comes from an isomorphism in $\mathbf{Set}^{\text{op}}$ (Lemma 2.3); likewise the unitors. Their pentagon and triangle are the pentagon and triangle of $\odot$ at representables. So $(\mathbf{Set}, \star, I)$ is monoidal (a monoidal structure on $\mathcal{C}$ and on $\mathcal{C}^{\text{op}}$ are the same data, with the coherence isomorphisms inverted).

Now $\odot$ and $\odot_\star$ are both bifunctors $\mathbf{Cont}^2 \rightarrow \mathbf{Cont}$ preserving coproducts in each variable — the first by (D1), the second by Proposition 4.2 — and they agree on representables by construction. **Cont** has small coproducts, so Lemma 2.5 applies with $\mathcal{E} = \mathbf{Cont}$: with $n = 2$ it makes $\odot$ and $\odot_\star$ canonically isomorphic. Their associators are both natural transformations between functors $\mathbf{Cont}^3 \rightarrow \mathbf{Cont}$ that preserve coproducts in each variable, and on representables both restrict to the associator of $\star$ — for $\odot$ by the construction just given, for $\odot_\star$ by Proposition 4.2. By Lemma 2.5 with $n = 3$, a natural transformation between two such functors is determined by its restriction to representables, so the two associators agree under the isomorphism; with $n = 1$, so do the unitors. Hence $\odot \cong \odot_\star$ is a monoidal isomorphism.

Fully faithful. A monoidal isomorphism $\varphi: \odot_\star \cong \odot_{\star'}$ over $\text{id}_{\mathbf{Cont}}$ restricts at representables to $y^{A \star B} \cong y^{A \star' B}$, hence (Lemma 2.3) to a natural isomorphism $\star \cong \star'$, monoidal because φ is. Conversely a monoidal isomorphism $\star \cong \star'$ extends by Lemma 2.5. The two operations are mutually inverse: restriction after extension is the identity on the nose, and extension after restriction is the identity by the uniqueness half of Lemma 2.5. $\square$

Remark 4.8. (D1) and (D2) are exactly what Lemma 2.5 needs: (D1) says the tensor is determined by its restriction to representables; (D2) says that restriction lands in $\mathbf{Set}^{\text{op}}$ rather than merely in **Cont**. Drop (D2) and one gets not a monoidal structure on **Set** but a promonoidal-style structure valued in **Cont** — a strictly larger family, which we do not classify here.

	(D1) coproducts, each variable	(D2) representables closed	Day tensor?
$\times$	yes	yes	yes, of $(+, \emptyset)$
$\otimes$	yes	yes	yes, of $(\times, 1)$
$+$	no — $p + 0 \cong p \not\cong 0$	no — $y^A + y^B$ has two shapes	no
$\triangleleft$	no in the right variable ²	yes — $y^A \triangleleft y^B = y^{A \times B}$	no

Table 1: The four, sorted by Theorem A’s two conditions.

4.2 Where the four sit

The coproduct fails both conditions, and fails (D1) for a triviality. *The sequential operator is the near miss.* It satisfies (D2) exactly, and its restriction to representables is $(A, B) \mapsto A \times B$ with unit 1 — the *same restriction as the Dirichlet tensor*. Only (D1), and only in the right variable, keeps it out of the family.

That is not a coincidence. Section 7 shows it is a theorem: $\otimes$ is what you get when you force $\triangleleft$ to satisfy (D1).

5 Pointwiseness, and the theorem that explains a dead end

Definition 5.1 (Pointwise). A monoidal structure $\odot$ on **Cont** is *pointwise* if there is an isomorphism $\theta_{p,q}: p \odot q \cong p \times q$, natural in p and q . [MacBeth]

Because $\llbracket - \rrbracket$ is fully faithful and $\llbracket p \times q \rrbracket = \llbracket p \rrbracket \times \llbracket q \rrbracket$, this says exactly that $\llbracket p \odot q \rrbracket X \cong \llbracket p \rrbracket X \times \llbracket q \rrbracket X$ naturally in p, q and X — i.e. that $\llbracket - \rrbracket: (\mathbf{Cont}, \odot) \rightarrow ([\mathbf{Set}, \mathbf{Set}], \times_{\text{ptwise}})$ is strong monoidal.

5.1 The unit already decides it — sometimes

Proposition 5.2. *If $\odot$ is pointwise then its unit is $y^\emptyset$, the terminal container.* [MacBeth]

Proof. Naturality gives $\llbracket J \rrbracket X \times \llbracket p \rrbracket X \cong \llbracket J \odot p \rrbracket X \cong \llbracket p \rrbracket X$ for all p, X . Take $p = y$, so $\llbracket p \rrbracket X = X$: then $\llbracket J \rrbracket X \times X \cong X$ for all X . At $X = 2$ this forces $|\llbracket J \rrbracket 2| = 1$. Writing $\llbracket J \rrbracket 2 = \sum_{s \in S_J} 2^{J[s]}$, a sum of positive cardinals equal to 1 forces $|S_J| = 1$ and $2^{|J[*]|} = 1$, i.e. $J[*] = \emptyset$. So $J = y^\emptyset = 1$. □

Corollary 5.3. *The Dirichlet tensor is not pointwise: its unit is $y \neq 1$.* [MacBeth]

That is a one-line proof of a fact we had previously only witnessed by the example $y^3 \otimes y^3 = y^9 \neq y^6$. But the unit is a *blunt* instrument: §6 exhibits a proper class of non-pointwise Day tensors that all share the product’s unit. Something sharper is needed, and the sharper thing is a rigidity property of the coproduct.

5.2 The rigidity of the coproduct

Lemma 5.4 (Rigidity). *Let $\alpha: (A + B) + C \Rightarrow A + (B + C)$ be any natural transformation of functors $\mathbf{Set}^3 \rightarrow \mathbf{Set}$. Then α is the canonical associator. Likewise the only natural $\emptyset + A \Rightarrow A$ and $A + \emptyset \Rightarrow A$ are the canonical unitors. Consequently $(+, \emptyset)$ admits exactly one monoidal structure on **Set**, and its coherence data are forced before coherence is imposed.* [MacBeth; likely folklore]

² $(y + 1) \triangleleft (1 + 0) \cong 2$ but $((y + 1) \triangleleft 1) + ((y + 1) \triangleleft 0) \cong 3$ [3, Ex. 6.56]. In the left variable $\triangleleft$ does preserve coproducts.

Proof. A map out of $(A + B) + C$ is precisely its three components on A, B, C .

The *A-component* is a family $u_{A,B,C}: A \rightarrow A + B + C$ natural in all three variables. Naturality in B and C along the unique maps $\emptyset \rightarrow B, \emptyset \rightarrow C$ gives $u_{A,B,C} = (\text{id}_A + !_B + !_C) \circ u_{A,\emptyset,\emptyset}$. Now $A + \emptyset + \emptyset \cong A$ canonically and naturally, so $u_{A,\emptyset,\emptyset}$ is a natural endomorphism of $\text{id}_{\mathbf{Set}}$, and the only one is the identity: for $a \in A$, name it $\bar{a}: 1 \rightarrow A$; naturality against $\bar{a}$ gives $\eta_A(a) = \eta_A(\bar{a}(*)) = \bar{a}(\eta_1(*)) = a$, since η_1 is the identity on 1. Hence $u_{A,B,C} = \text{inl}$.

The *B-component* is a family $v_{A,B,C}: B \rightarrow A + B + C$. Setting $A = C = \emptyset$ and running the same argument makes $v_{\emptyset,B,\emptyset}$ the canonical isomorphism; naturality in A and C along $\emptyset \rightarrow A, \emptyset \rightarrow C$ then makes $v_{A,B,C}$ the middle injection. Symmetrically for C . So α is canonical.

Unitors. A natural $\lambda_A: \emptyset + A \rightarrow A$ has an $\emptyset$ -component (unique) and an A -component, a natural endomorphism of $\text{id}_{\mathbf{Set}}$, hence the identity. $\square$

5.3 Theorem B, and the removal of its hypothesis

Theorem 5.5 (B: the product is the unique pointwise Day tensor). *For a monoidal structure $(\star, I)$ on $\mathbf{Set}$, the following are equivalent:*

- (1) $\odot_\star$ is pointwise;
- (2) $\star \cong +$ as bifunctors $\mathbf{Set} \times \mathbf{Set} \rightarrow \mathbf{Set}$;
- (3) $(\star, I) \cong (+, \emptyset)$ as monoidal structures on $\mathbf{Set}$;
- (4) $\odot_\star \cong \times$ as monoidal structures on $\mathbf{Cont}$.

[MacBeth]

Proof. (1) $\Rightarrow$ (2). Restrict θ to representables. By Proposition 4.2, $y^A \odot_\star y^B = y^{A \star B}$; and $y^A \times y^B = y^{A+B}$ (one shape, positions $A+B$). So θ gives $y^{A \star B} \cong y^{A+B}$, natural in $A, B \in \mathbf{Set}^{\text{op}}$. Since $y^{(-)}$ is fully faithful (Lemma 2.3), this isomorphism comes from a natural isomorphism of the bifunctors $\star$ and $+$ on $(\mathbf{Set}^{\text{op}})^2$; taking opposites, $\star \cong +$ on $\mathbf{Set}^2$.

(2) $\Rightarrow$ (3). Let $\varphi: \star \cong +$ be a natural isomorphism of bifunctors. The left unitor of $\star$ gives $I \star A \cong A$; composing with φ , $I + A \cong A$ for every set A . At $A = 1$ this gives $|I| + 1 = 1$, so $I = \emptyset$. Transport the monoidal structure $(\star, I, \alpha, \lambda, \rho)$ along φ and $I \cong \emptyset$: the result is a monoidal structure whose tensor is literally $+$, whose unit is literally $\emptyset$, and whose coherence data are natural — hence, by Lemma 5.4, *canonical*. So the transported structure is $(+, \emptyset)$ on the nose, i.e. $(\varphi, I \cong \emptyset)$ is a monoidal isomorphism.

(3) $\Rightarrow$ (4). The fully faithful half of Theorem 4.7: a monoidal isomorphism $(\star, I) \cong (+, \emptyset)$ extends to a monoidal isomorphism $\odot_\star \cong \odot_{(+, \emptyset)} = \times$ (Example 4.3).

(4) $\Rightarrow$ (1). A monoidal isomorphism is in particular a natural isomorphism. $\square$

Theorem 5.6 (B^+ : the Day hypothesis is not needed). *Let $(\odot, J)$ be any monoidal structure on $\mathbf{Cont}$. Then $\odot$ is pointwise if and only if $(\odot, J) \cong (\times, 1)$. That is: **the categorical product is the unique pointwise monoidal structure on $\mathbf{Cont}$.***

[MacBeth]

Proof. $(\Leftarrow)$ is trivial. $(\Rightarrow)$ Pointwise says $\odot \cong \times$ as bifunctors. Both (D1) and (D2) are properties of the bifunctor alone and are invariant under natural isomorphism of bifunctors. The product satisfies both: (D1) because $\mathbf{Cont}$ is distributive ($(p + q) \times r \cong p \times r + q \times r$ and $0 \times r \cong 0$, immediate from the shape formula $S_p \times S_q$), and (D2) because $y^A \times y^B = y^{A+B}$. Hence $\odot$ satisfies (D1) and (D2), i.e. $(\odot, J)$ is convolutional. By Theorem 4.7, $(\odot, J) \cong (\odot_\star, y^I)$ monoidally for some monoidal $(\star, I)$ on $\mathbf{Set}$; then $\odot_\star$ is pointwise, so by Theorem 5.5 $\odot_\star \cong \times$ monoidally, and therefore $\odot \cong \times$ monoidally. $\square$

So pointwiseness is not a condition one imposes *within* the Day family and then discovers is rare. It is a condition on monoidal structures on $\mathbf{Cont}$ at large, and it holds for the product and nothing else. The Day family is simply where one can *see* this, because Theorem A makes the family surveyable.

Corollary 5.7 (Why the strictness dead end was forced). $\llbracket - \rrbracket : (\mathbf{Cont}, \odot) \rightarrow ([\mathbf{Set}, \mathbf{Set}], \times_{ptwise})$ is strong monoidal if and only if $\odot$ is the cartesian structure. In particular no reformulation of $\otimes$ could ever have made $\llbracket - \rrbracket$ monoidal into the pointwise product. [MacBeth]

This retro-explains the erratum of §1. The failure was structural, not a bookkeeping error: there was no version of that claim that could have been true. (What *is* true is that $\llbracket - \rrbracket$ is strong monoidal from $(\mathbf{Cont}, \otimes, y)$ to $([\mathbf{Set}, \mathbf{Set}], \text{Day-of-}\times, \text{id})$ — into the Dirichlet convolution, not into the pointwise product. The comparison there is the co-Yoneda isomorphism, and it is not an identity.)

5.4 An effective test

Theorem B is an existence statement; to *use* it one wants to inspect a given $\star$ and decide. That is what the next definition does.

Definition 5.8 (Comparison map). Let $(\star, I)$ be monoidal on $\mathbf{Set}$ with $I \cong \emptyset$. The *comparison map* $\kappa_{A,B} : A + B \rightarrow A \star B$ is the copairing of

$$A \cong A \star \emptyset \xrightarrow{A \star !_B} A \star B \quad \text{and} \quad B \cong \emptyset \star B \xrightarrow{!_A \star B} A \star B,$$

using the unitors and functoriality of $\star$ along $! : \emptyset \rightarrow X$. It is natural in A and B . [MacBeth]

Theorem 5.9 (B' : test form). $\odot_\star$ is pointwise *iff* $I \cong \emptyset$ and $\kappa_{A,B}$ is a bijection for all A, B . [MacBeth]

Proof. $(\Leftarrow)$ κ is then a natural isomorphism $+$ $\cong$ $\star$, so Theorem 5.5(2) applies.

$(\Rightarrow)$ Suppose $\odot_\star$ is pointwise. Then $I \cong \emptyset$ (Proposition 5.2, since $J_\star = y^I$), so κ is defined; and by Theorem 5.5 there is a *monoidal* natural isomorphism $(\varphi, \iota) : (\star, I) \rightarrow (+, \emptyset)$. We claim $\varphi \circ \kappa = \text{id}_{A+B}$, whence $\kappa = \varphi^{-1}$ is a bijection. It suffices to check the two components. On the A -summand, $\kappa|_A$ is $A \xrightarrow{(\rho^\star)^{-1}} A \star I \xrightarrow{A \star !_B} A \star B$. Naturality of φ in the second variable gives $\varphi_{A,B} \circ (A \star !_B) = (A + !_B) \circ \varphi_{A,I}$; and compatibility of (φ, ι) with the right unitors gives $\varphi_{A,I} \circ (\rho^\star)^{-1} = (\rho^+)^{-1}$, the canonical $A \cong A + \emptyset$. Composing, $\varphi_{A,B} \circ \kappa|_A = (A + !_B) \circ (A \cong A + \emptyset) = \text{inl}$. Symmetrically, using the left unitor, $\varphi_{A,B} \circ \kappa|_B = \text{inr}$. So $\varphi \circ \kappa = [\text{inl}, \text{inr}] = \text{id}$. $\square$

The content of the check is worth extracting: κ is built out of unitors and functoriality alone, so *any* monoidal isomorphism carries $\kappa^\star$ to κ^+ , and κ^+ is visibly the identity. Since κ is computable straight from the definition of $\star$, Theorem B' is what makes Theorem B usable. We now run it on a proper class.

6 The family is a proper class, and the unit sees nothing

If the product were distinguished among Day tensors by something cheap — its unit, say, or semicartesianness — then Theorem B would be a curiosity. It is not, and the following family is why.

Definition 6.1 ($\vee_S$). For a set S , define on $\mathbf{Set}$:

$$A \vee_S B := A + (A \times S \times B) + B, \quad \text{unit } \emptyset.$$

[Cited: Spivak [4, eq. (9)], who credits R. Garner; the case $S = 1$ is Niu–Spivak [3, Ex. 3.82]]

Read $A \vee_S B$ as “an A , or a B , or both with an S -labelled link”. Spivak asserts that this is monoidal; Niu–Spivak set the $S = 1$ case as an exercise in *associativity*. We need genuine *coherence*, so we prove it — and §6.1 shows the distinction is not pedantry.

Proposition 6.2 ($\vee_S$ is monoidal). $(\vee_S, \emptyset)$ is a monoidal structure on **Set**, and $\vee_\emptyset \cong +$. [MacBeth (coherence); the structure is Cited: Spivak [4, eq. (9)]]

Proof (coherence by normal form). For $n \geq 1$ define

$$N(X_1, \dots, X_n) := \coprod_{\emptyset \neq K \subseteq [n]} \left(\prod_{i \in K} X_i \right) \times S^{|K|-1}.$$

Read an element as a *word*: choose a nonempty $K = \{i_1 < \dots < i_k\} \subseteq [n]$, a letter $x_{i_j} \in X_{i_j}$ for each, and a separator $s_j \in S$ between each *consecutive* pair,

$$x_{i_1} \cdot s_1 \cdot x_{i_2} \cdot s_2 \cdots s_{k-1} \cdot x_{i_k}.$$

N is functorial in each variable; $N(A) = A$, and $N(A, B) = A \sqcup (A \times S \times B) \sqcup B = A \vee_S B$, the three summands being $K = \{1\}, \{1, 2\}, \{2\}$.

Splitting. For each $1 \leq m < n$ there is a bijection, natural in all X_i ,

$$c_m: N(X_1, \dots, X_m) \vee_S N(X_{m+1}, \dots, X_n) \longrightarrow N(X_1, \dots, X_n),$$

given by concatenation. Indeed, unfolding the source it is $N(X_{\leq m}) \sqcup (N(X_{\leq m}) \times S \times N(X_{> m})) \sqcup N(X_{> m})$, and every word w of $N(X_1, \dots, X_n)$ falls into exactly one of three cases: $\text{supp}(w) \subseteq [1, m]$; $\text{supp}(w) \subseteq [m+1, n]$; or $\text{supp}(w)$ meets both. In the third case w has a *unique* separator straddling the cut $m \mid m+1$ — separators sit between consecutive *chosen* indices, and exactly one such pair crosses the cut — so w factors uniquely as $w_{\leq} \cdot s \cdot w_{>}$, which is precisely the middle summand. The inverse is “cut at the straddling separator”.

Normal form. For a binary bracketing β of $X_1, \dots, X_n$ define $\text{can}_\beta: \beta(X_1, \dots, X_n) \rightarrow N(X_1, \dots, X_n)$ by induction: $\text{can}_{(1)} = \text{id}$, and if β splits at m into β_1, β_2 then $\text{can}_\beta := c_m \circ (\text{can}_{\beta_1} \vee_S \text{can}_{\beta_2})$. Each can_β is a natural bijection. Define the associator at any node by $\alpha := \text{can}_{\beta'}^{-1} \circ \text{can}_\beta$. This is consistent with whiskering: if $\gamma = \text{can}_{\beta'_2}^{-1} \circ \text{can}_{\beta_2}$ then

$$\text{can}_{(\beta_1, \beta'_2)} \circ (\text{id} \vee_S \gamma) = c_m \circ (\text{can}_{\beta_1} \vee_S (\text{can}_{\beta'_2} \circ \gamma)) = c_m \circ (\text{can}_{\beta_1} \vee_S \text{can}_{\beta_2}) = \text{can}_{(\beta_1, \beta_2)},$$

and symmetrically on the other side. Consequently *every* morphism built by composing and whiskering associators, from a bracketing β to a bracketing β' , equals $\text{can}_{\beta'}^{-1} \circ \text{can}_\beta$ — so any two such morphisms $\beta \rightarrow \beta'$ are equal. The pentagon is exactly such a pair, so it commutes. (This proves the pentagon rather than assuming it.)

Unitors and triangle. $\emptyset \vee_S B = \emptyset + (\emptyset \times S \times B) + B \cong B$, and symmetrically. Substituting $X_i = \emptyset$ into N deletes every word whose support contains i , leaving N of the remaining arguments; so the unitors are can-compatible and the triangle commutes by the same argument. Finally $\vee_\emptyset$ has $A \times \emptyset \times B = \emptyset$, so $A \vee_\emptyset B \cong A + B$. $\square$

Remark 6.3 (We do *not* claim $\vee_S$ is symmetric). There is an evident candidate braiding, and Spivak’s construction is stated for symmetric monoidal structures on **Set**. But we have not verified the hexagon, and we decline to assume it. The reason is specific, not fastidious: a braiding permutes the arguments, and a permutation changes which pairs of chosen indices are *consecutive* — so it must re-match the S -separators, and re-matching separators is exactly where coherence fails (§6.1). Nothing below needs symmetry.

Definition 6.4. $p \triangleright_S q := \odot_{\vee_S}(p, q) = \sum_{(s,t)} y^{p[s] \vee_S q[t]}$. [Cited: Spivak [4, eq. (12)]]

Theorem 6.5 (The unit is blind). *For every set S , $\triangleright_S$ is a convolutional monoidal structure on **Cont** with unit $y^\emptyset = 1$, the terminal object — the same unit as the categorical product. The $\triangleright_S$ are pairwise non-isomorphic. And*

$$\triangleright_S \text{ is pointwise } \iff S = \emptyset \iff \triangleright_S = \times.$$

[MacBeth]

Proof. Convolutionality and the unit: Propositions 4.2 and 6.2. Pairwise non-isomorphic: by Theorem 4.7 it suffices that the $\vee_S$ be pairwise non-isomorphic, and $|1 \vee_S 1| = 1 + |S| + 1 = |S| + 2$ separates them. Pointwise: apply Theorem 5.9. The unit is already initial, and

$$\kappa_{A,B}: A + B \hookrightarrow A + (A \times S \times B) + B$$

is the inclusion of the two outer summands (compute: $A \cong A \vee_S \emptyset$ is the first summand, and functoriality along $\emptyset \rightarrow B$ includes it into the first summand of $A \vee_S B$; symmetrically for B). So κ is a bijection for all A, B iff $A \times S \times B = \emptyset$ for all A, B , iff $S = \emptyset$. And $\triangleright_\emptyset = \odot_{\vee_\emptyset} = \odot_+ = \times$ by Proposition 6.2 and Example 4.3. $\square$

Corollary 6.6 (The sharp form of “unique”). **Cont** carries a proper class of pairwise non-isomorphic convolutional monoidal structures, all of which have the terminal object as unit — and exactly one of them, the case $S = \emptyset$, is the categorical product. [MacBeth]

So the product is *not* singled out among Day tensors by its unit, nor by coproduct preservation, nor by semicartesianness (every $\triangleright_S$ is semicartesian). The property that singles it out is precisely pointwiseness, and Theorem B' is what detects it. The obstruction is always the same object: the middle summand $A \times S \times B$, the “both, linked” words that $+$ cannot see.

6.1 A negative control, and what it says about the literature

All of the above was checked by machine on finite containers (see §10). One item of that verification deserves to be in the text, because it changes how one should read the sources.

We constructed a *label-swapping variant* α' of the $\vee_S$ associator: it is a natural bijection, it matches cardinalities, and it *is associative*. It nonetheless **fails the pentagon** — first at $|A| = |B| = |C| = |D| = 1$ with $|S| = 2$, where the two legs send

$$(\text{both}, (\text{both}, (\text{both}, (a, k_0, b)), k_1, c), k_1, d)$$

to words differing in which separator label sits where. Two consequences.

1. The pentagon test has *teeth*: our PASS on the true associator is not vacuous. Note it takes $|S| \geq 2$ to see the difference — a test suite with $|S| \leq 1$ would have blessed the impostor.
2. Proposition 6.2 was not pedantry. Spivak [4, eq. (9)] asserts monoidality; Niu–Spivak [3, Ex. 3.82] ask the reader for a bijection $(A \vee B) \vee C \cong A \vee (B \vee C)$ and unitality, and nothing more. The negative control exhibits a natural, associative bijection that is *not* a monoidal associator. Associativity-as-a-bijection genuinely does not suffice, so the coherence proof — on which Corollary 6.6 depends — fills a real gap rather than restating a source.

The moral generalises past this chapter: *when a construction is asserted to be monoidal, run a negative control*. Build a plausible impostor associator and check that your test rejects it. If it does not, your test proved nothing.

7 How the four relate: the comparator is a coreflection counit

We return to the near miss of Table 1. The sequential operator satisfies (D2) and fails only (D1), and only in the right variable. Its restriction to representables — $(A, B) \mapsto A \times B$, unit 1 — is *the same* as the Dirichlet tensor’s. By Theorem A that restriction determines a unique convolutional tensor. So there is exactly one candidate for “the coproduct-preserving shadow of $\triangleleft$ ”, and it is $\otimes$.

The literature knows a map between them. Niu–Spivak [3, Ex. 6.85] and Spivak [4, eq. (32)] record a natural transformation $o_{p,q}: p \otimes q \rightarrow p \triangleleft q$ — the *comparator* — usually obtained by feeding y into the duoidal interchanger $(p \triangleleft p') \otimes (q \triangleleft q') \rightarrow (p \otimes q) \triangleleft (p' \otimes q')$ [5]. That derivation tells you the map exists. It does not tell you what it *is*. Theorem C does.

Theorem 7.1 (C). *Write $J := y^{(-)}: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Cont}$ for the representable embedding, and fix $p \in \mathbf{Cont}$. Put $\Phi_p := (p \triangleleft -)$ and $\Psi_p := (p \otimes -)$. Then:*

- (1) $p \triangleleft y^B \cong p \otimes y^B$ naturally: $\Phi_p \circ J \cong \Psi_p \circ J$.
- (2) $\Psi_p \cong \text{Lan}_J(\Phi_p \circ J)$. That is, $p \otimes (-)$ **is the left Kan extension of $p \triangleleft (-)$ along the representable embedding** — equivalently, the free coproduct-preserving functor agreeing with $p \triangleleft (-)$ on representables.
- (3) The counit $o_{p,q}: p \otimes q \rightarrow p \triangleleft q$ is the **counit of the adjunction** $\text{Lan}_J \dashv \text{res}_J$ at Φ_p .
- (4) Hence $p \otimes (-)$ is the coreflection of $p \triangleleft (-)$ into coproduct-preserving endofunctors: for every coproduct-preserving $F: \mathbf{Cont} \rightarrow \mathbf{Cont}$, postcomposition with the comparator is a bijection

$$\text{Nat}(F, p \otimes -) \cong \text{Nat}(F, p \triangleleft -).$$

$p \otimes (-)$ is the terminal coproduct-preserving approximation to $p \triangleleft (-)$.

[MacBeth; part (1) is Cited: Spivak [4, eq. (33)], Niu–Spivak [3, Ex. 6.84]. The comparator itself is Cited: Niu–Spivak [3, Ex. 6.85]; MacBeth, Lean-verified: *Dirichlet.lean (Container.dirToSeq)*]

Proof. (1) $p \triangleleft y^B$ has shapes $\sum_s (p[s] \rightarrow 1) \cong S_p$ and positions $\sum_{i \in p[s]} B = p[s] \times B$; $p \otimes y^B$ has shapes $S_p \times 1 \cong S_p$ and positions $p[s] \times B$. The evident map is the identity on both.

(2) By Lemma 2.6, left Kan extension along J is “extend by coproducts”: $(\text{Lan}_J G)(q) = \sum_{t \in S_q} G(q[t])$. Taking $G = \Phi_p \circ J$, i.e. $G(B) = p \triangleleft y^B \cong p \otimes y^B$ by (1),

$$(\text{Lan}_J(\Phi_p \circ J))(q) = \sum_{t \in S_q} (p \otimes y^{q[t]}) = \sum_{t \in S_q} \sum_{s \in S_p} y^{p[s] \times q[t]} = \sum_{(s,t)} y^{p[s] \times q[t]} = p \otimes q.$$

(3) The counit $\varepsilon: \text{Lan}_J(\Phi_p \circ J) \Rightarrow \Phi_p$ is, at q , the map induced on the coproduct $\sum_{t \in S_q} (p \triangleleft y^{q[t]})$ by the family $p \triangleleft \iota_t$, where $\iota_t: y^{q[t]} \rightarrow q$ is the t -th coproduct injection (shape $*$ maps to t , positions the identity on $q[t]$). Compute $p \triangleleft \iota_t$: a shape of $p \triangleleft y^{q[t]}$ is $(s, f: p[s] \rightarrow 1)$, i.e. just s , and it is sent to (s, const_t) ; on positions it is the identity of $\sum_{i \in p[s]} q[t]$. Under the identification of (2) the source is $p \otimes q$, with shape (s, t) and positions $p[s] \times q[t]$. So

$$\varepsilon: \text{shapes } (s, t) \mapsto (s, \text{const}_t), \quad \text{positions id: } p[s] \times q[t] = \sum_{i \in p[s]} q[t],$$

which is precisely the comparator.

(4) $\text{Lan}_J \dashv \text{res}_J$ (Lemma 2.6) gives $\text{Nat}(\text{Lan}_J G, \Phi) \cong \text{Nat}(G, \Phi \circ J)$ for every Φ — in particular for $\Phi = \Phi_p$, which does not preserve coproducts. Every coproduct-preserving F satisfies $F \cong \text{Lan}_J(F \circ J)$ (Lemma 2.5), and Lan_J is fully faithful, since the unit of the adjunction is invertible: $(\text{Lan}_J G)(y^A) = \sum_{t \in 1} G(A) = G(A)$ by the formula of Lemma 2.6, y^A having a single shape with positions A . Hence

$$\text{Nat}(F, \Phi_p) \cong \text{Nat}(F \circ J, \Phi_p \circ J) \cong \text{Nat}(\text{Lan}_J(F \circ J), \text{Lan}_J(\Phi_p \circ J)) \cong \text{Nat}(F, \Psi_p),$$

and unwinding, the composite bijection is postcomposition with the counit $\varepsilon = o_{p,-}$. $\square$

Corollary 7.2 (What the comparator *is*). *The Dirichlet tensor is the Day-ification of the sequential operator: $\otimes$ is the unique convolutional tensor agreeing with $\triangleleft$ on representables (Theorem 4.7), and the comparator is the universal map out of it. Two of the three standard facts about the comparator become corollaries rather than computations:*

- it exists at all — *it is a counit (the duoidal derivation is a consequence of what it is, not the reason it is there)*;
- it is invertible when q is representable [4, eq. (33)] — *because the counit of $\text{Lan}_J \dashv \text{res}_J$ is invertible on the image of J , J being fully faithful. No computation needed.*

[MacBeth]

The third standard fact — compatibility with the associators — we verify by hand rather than appeal to a general monoidal-Kan-extension principle we have not checked.

Proposition 7.3 (Coherence of the comparator). *The family $o_{p,q}: p \otimes q \rightarrow p \triangleleft q$ makes $\text{id}_{\mathbf{Cont}}$ a lax monoidal functor $(\mathbf{Cont}, \triangleleft, y) \rightarrow (\mathbf{Cont}, \otimes, y)$: it commutes with the associators and the unitors.* [Content Cited: Niu–Spivak [3, Ex. 6.85, Prop. 6.87], Spivak [4, eq. (32)]; proof MacBeth]

Proof. Both composites $(a \otimes b) \otimes c \rightarrow a \triangleleft (b \triangleleft c)$ have source shapes (s_a, s_b, s_c) and source positions $a[s_a] \times b[s_b] \times c[s_c]$, and every map involved is the identity on positions (each o is, by Theorem 7.1(3); the associators are canonical rebracketings). So it suffices to compare shape maps. Along the top, $o \otimes 1$ sends $((s_a, s_b), s_c) \mapsto ((s_a, \lambda i. s_b), s_c)$; then o gives $((s_a, \lambda i. s_b), \lambda(i, j). s_c)$; then the $\triangleleft$ -associator (Proposition 3.4) yields $(s_a, \lambda i. (s_b, \lambda j. s_c))$. Along the bottom, the $\otimes$ -associator gives $(s_a, (s_b, s_c))$; then $1 \otimes o$ gives $(s_a, (s_b, \lambda j. s_c))$; then o gives $(s_a, \lambda i. (s_b, \lambda j. s_c))$ — the same shape. For the unitors, $y \otimes q = q = y \triangleleft q$ and $p \otimes y = p = p \triangleleft y$ with o the identity in both cases (the shape map $(s, *) \mapsto (s, !)$ is the identity, $p[s] \rightarrow 1$ having a unique element). $\square$

Remark 7.4 (A convention clash, resolved). Niu–Spivak [3, Ex. 6.85] write the source as $\otimes$ and the target as $\triangleleft$; Spivak [4, eq. (32)] writes it the other way. Only the latter type-checks under the usual definition of *lax*: a lax $F: (\mathcal{C}, \odot_{\mathcal{C}}) \rightarrow (\mathcal{D}, \odot_{\mathcal{D}})$ has structure map $Fa \odot_{\mathcal{D}} Fb \rightarrow F(a \odot_{\mathcal{C}} b)$, which for $F = \text{id}$, source $\triangleleft$, target $\otimes$ reads $p \otimes q \rightarrow p \triangleleft q$. The mathematical content — “ o commutes with associators and unitors” — is the same either way, and is what Proposition 7.3 proves.

7.1 Exactly where the comparator is invertible

Proposition 7.5 (Trichotomy). *The position component of $o_{p,q}$ is always the identity, so $o_{p,q}$ is an isomorphism iff its shape map is a bijection. That shape map is the coproduct over $s \in S_p$ of*

$$c_s: S_q \longrightarrow S_q^{p[s]}, \quad t \mapsto \text{const}_t,$$

and therefore:

$o_{p,q}$ is an isomorphism iff ...	
q representable ($ S_q = 1$)	always
$q = 0$ ($S_q = \emptyset$)	every $p[s] \neq \emptyset$
$ S_q \geq 2$	p is linear (every $p[s]$ a singleton)

[MacBeth; the sufficient direction is Cited: Spivak [4, eq. (33)]]

Proof. If $|p[s]| = 1$ then $S_q^{p[s]} \cong S_q$ and c_s is the identity. If $p[s] = \emptyset$ then $S_q^{\emptyset} = 1$ and $c_s: S_q \rightarrow 1$ is bijective iff $|S_q| = 1$. If $|p[s]| \geq 2$ then c_s is surjective iff every $f: p[s] \rightarrow S_q$ is constant, i.e. iff $|S_q| \leq 1$; and if $S_q = \emptyset$ then $S_q^{p[s]} = \emptyset$ too and c_s is a bijection. Assembling the cases gives the table. $\square$

The reading of Proposition 7.5 is the sentence this chapter has been walking towards:

Coproduct-preservation in the second variable is precisely non-dependence of the inner shape on the outer position. A shape of $p \triangleleft q$ is $(s, f: p[s] \rightarrow S_q)$ — a choice of inner shape for each position of the outer one. A shape of $p \otimes q$ is (s, t) : the case where f is constant. A linear p has one position per shape, so there is nothing for f to depend on — and there the two agree exactly.

Proposition 7.6 (The comparitor is neither monic nor epic). *Take $p = 1 = y^\emptyset$. Then $1 \otimes q = \sum_{t \in S_q} y^{\emptyset \times q[t]} = S_q \cdot y^\emptyset$, the constant container at S_q , while $1 \triangleleft q = \llbracket 1 \rrbracket \circ \llbracket q \rrbracket = 1$. So $o_{1,q}: S_q \cdot 1 \rightarrow 1$ collapses everything, and is not injective on shapes as soon as $|S_q| \geq 2$. And for $p = y^2$, $q = 2$ the shape map $S_p \times S_q \rightarrow \sum_s S_q^{p[s]}$ is $2 \rightarrow 4$, not surjective. [MacBeth, Lean-verified: `Dirichlet.lean (dirToSeq_not_surjective)`]*

Both failures are informative, and in *opposite* directions.

- **Not surjective:** $\triangleleft$ has genuinely dependent shapes that $\otimes$ cannot reach. This is the dependency that $\triangleleft$ has and $\otimes$ lacks.
- **Not injective:** when an outer shape has *no* positions, $\triangleleft$ cannot see the inner choice at all — there is nowhere to put it — but $\otimes$ still records it. The comparitor forgets it.

So the failure of o to be an isomorphism measures *dependency* in one direction and *vacuity* in the other. That is why $\otimes$ is a coreflection of $\triangleleft$ and not a subobject of it.

The sentence to remember. $\otimes$ is $\triangleleft$ with the dependency switched off — not by fiat, but universally: $p \otimes (-)$ is the terminal coproduct-preserving approximation to $p \triangleleft (-)$, and the comparitor is its counit. It fails to be surjective exactly where $\triangleleft$ is dependent, and fails to be injective exactly where $\triangleleft$ is vacuous.

8 Closing the structures

A monoidal category invites one question above all others: is the tensor *closed*? For $\times$ and $+$ the answers are the classical ones — **Cont** has finite products and coproducts, and $(\mathbf{Cont}, \times, 1)$ is cartesian closed (below). This section records the two answers that are specific to **Cont**: the Dirichlet tensor is closed, with an internal hom whose *shapes are the morphisms*; and we can say exactly what closure of a general convolutional tensor looks like. Almost everything here is cited. What is ours is a Lean certification, and one uniform formula.

8.1 The Dirichlet closure

Definition 8.1 (Internal hom for $\otimes$). For containers q, r define

$$[q, r] := \left(\mathbf{Cont}(q, r), (u, \varphi) \mapsto \sum_{t \in S_q} r[ut] \right) = \sum_{\varphi: q \rightarrow r} y^{\sum_{t \in S_q} r[\varphi_1 t]}.$$

Its shapes are the container morphisms $q \rightarrow r$; over a shape (u, φ) (shape map $u: S_q \rightarrow S_r$, position map φ) the positions are the pairs (t, i) with $t \in S_q$ and $i \in r[ut]$. [Cited: Niu–Spivak [3, Ex. 4.78, eq. (4.79)]; Spivak [4, eq. (44)]; Spivak–Srinivasan [6, Prop. 2.8]]

Theorem 8.2 ($\otimes$ is closed). *For each q the functor $(- \otimes q)$ is left adjoint to $[q, -]$: there is a bijection*

$$\mathbf{Cont}(p \otimes q, r) \cong \mathbf{Cont}(p, [q, r]),$$

natural in p and in r . Hence $(\mathbf{Cont}, \otimes, y)$ is a (symmetric) closed monoidal category. [Cited: Niu–Spivak [3, Ex. 4.78]; MacBeth, Lean-verified: `DirichletClosed.lean (dirichlet_closure)`]

Proof (the transpose, in one breath). Unfold both sides and rearrange. A morphism $p \otimes q \rightarrow r$ is a shape map $U: S_p \times S_q \rightarrow S_r$ together with position maps $r[U(s, t)] \rightarrow p[s] \times q[t]$. Splitting each position map into its two components (Hom out of a fixed object preserves products) and applying the type-theoretic axiom of choice twice — once to pull the S_r -choice out over S_q , once to pull the whole package out over S_p — turns the datum into: for each $s \in S_p$, a morphism $q \rightarrow r$ (this is where the q -component of the position map becomes part of a *shape*) together with a map $(\sum_t r[u_s t]) \rightarrow p[s]$. That is exactly a morphism $p \rightarrow [q, r]$. The transpose is a bijection because each rearrangement is; it is natural in p (contravariantly, by precomposition) and in r (covariantly, by postcomposition). $\square$

The formula is not ours: it is Niu–Spivak’s Exercise 4.78 and Spivak’s “Dirichlet closure”. The derivation above reproduces it character for character. What *is* ours is a mechanisation.

Remark 8.3 (Machine-checked closure). The file `DirichletClosed.lean` constructs $[q, -]$, the currying bijection `dirCurry/dirUncurry`, *both* round-trip identities, and naturality in each variable. Every proof is `rfl`: the currying bijection is a rearrangement of a $\Sigma/\times$ telescope, and Lean’s definitional η for Σ , $\times$ and the morphism structure identifies the two round trips on the nose — no transports, no `funext`. The declaration `dirichlet_closure` depends on no axioms. To our knowledge this is the first machine-checked internal hom for the Dirichlet tensor. It matters for the book’s larger claim that these structures are *formally* under control: closure, the subtlest of the four questions, is here the cheapest. [MacBeth, Lean-verified: `DirichletClosed.lean`]

8.2 Closure of a general convolutional tensor

Theorem A makes it natural to ask for the closure of *every* Day tensor at once, not just $\otimes$. There is a uniform formula, and it explains a small mystery in the literature.

Remark 8.4 (A uniform closure formula). Let $(\star, I)$ be monoidal on **Set** and suppose each functor $R \star (-)$ is polynomial (preserves wide pullbacks); write $\odot_\star$ for the induced Day tensor (Definition 4.1). Then $\odot_\star$ is closed, with internal hom

$$[p, q]_\star = \prod_{I \in S_p} q \triangleleft (p[I] \star y), \quad \text{i.e.} \quad \llbracket [p, q]_\star \rrbracket A = \prod_{I \in S_p} \llbracket q \rrbracket (A \star p[I]),$$

the one-line derivation being

$$\mathbf{Cont}(r \odot_\star p, q) = \prod_{J \in S_r} \prod_{I \in S_p} \llbracket q \rrbracket (r[J] \star p[I]) = \mathbf{Cont}(r, [p, q]_\star).$$

For the three named cases this recovers exactly Spivak’s eqs. (38), (39), (40) [4] — the closures of $\times$, $\otimes$ and $\triangleright_S$ — which appear there *written side by side and never unified*; the specialisation $\star = \times$ recovers Definition 8.1. We grade this [computed]: the displayed identity is verified on finite containers (§10) and by the derivation above, but we have *not* checked the coherence of the adjunction in this generality. [MacBeth [computed]; the three instances are Cited: Spivak [4, eqs. (38)–(40)]]

Remark 8.5 (A tempting converse, and why we do not claim it). Remark 8.4 together with Theorem A suggests a clean statement — *the convolutional structures whose $\star$ has $R \star (-)$ polynomial are exactly the closed convolutional structures on **Cont*** — but the necessity direction (that a *closed* convolutional tensor forces $R \star (-)$ polynomial) we have not argued. We flag it [speculative] and leave it open. [Open [speculative]]

8.3 The sequential operator, and cartesian closure

The sequential operator is not symmetric, so “closed” bifurcates. It is not closed on the left, but it does have a *right coclosure*: a functor $\{q, -\}$ with $\mathbf{Cont}(p, q \triangleleft r) \cong \mathbf{Cont}(\{q, p\}, r)$, computed by Josh Meyers.

Theorem 8.6 (Coclosure of $\triangleleft$). *The substitution product $\triangleleft$ admits a right coclosure.* [Cited: Spivak [4, eqs. (68)–(69)]; Niu–Spivak [3, Prop. 6.57], attributed to J. Meyers]

Remark 8.7 (A naming clash to be aware of). The same isomorphism is called a *left* coclosure by Niu–Spivak [3, Prop. 6.57] and a *right* coclosure by Spivak [4, eq. (68)]. The variance convention differs, not the mathematics; a reader collating the two sources should expect the labels to disagree.

Finally, the cartesian structure. This is the closure that would matter for type theory, and the answer is a cautionary one.

Theorem 8.8 ($\mathbf{Cont}$ is cartesian closed but not locally cartesian closed). *($\mathbf{Cont}, \times, 1$) is cartesian closed, with exponential $\llbracket q^p \rrbracket \cong \prod_{s \in S_p} \llbracket q \rrbracket \circ (- + p[s])$. But $\mathbf{Cont}$ is not locally cartesian closed: its slices are not all cartesian closed, so $\mathbf{Cont}$ does not model extensional dependent type theory.* [Cited: Altenkirch–Levy–Staton [7]; the CCC structure is Niu–Spivak [3, Thm. 5.31, eq. (5.28)]]

The contrast with the Dirichlet closure is the point worth carrying away. $\mathbf{Cont}$ is closed for the tensor whose extension is *not* pointwise ($\otimes$), and *fails* to be closed in the strong (local) sense for the tensor whose extension *is* pointwise ($\times$). Closure and pointwiseness pull in opposite directions — which is a second reading of the erratum of §1: there is no bridge between “pointwise” and “closed”.

9 How the four interact: the duoidal category and its double comonoids

The final question a family of monoidal structures raises is how they fit *together*. For $\otimes$ and $\triangleleft$ the fit has a name.

Theorem 9.1 (The normal duoidal structure). *($\mathbf{Cont}, \otimes, \triangleleft$) is a normal duoidal category: $\otimes$ and $\triangleleft$ share the unit y , and there is an interchange*

$$\zeta: (p_1 \triangleleft p_2) \otimes (q_1 \triangleleft q_2) \longrightarrow (p_1 \otimes q_1) \triangleleft (p_2 \otimes q_2),$$

natural and coherent, making $\otimes$ a lax and $\triangleleft$ an oplax functor for one another. [Cited: Spivak–Srinivasan [5, eq. (29)]]

Feeding y into ζ recovers the comparator $o_{p,q}: p \otimes q \rightarrow p \triangleleft q$ of §7; that is how the comparator is usually produced. So the interchange is the *spine* of the interaction, and Theorem C already told us what sits on it: the comparator is the counit exhibiting $\otimes$ as the Day-ification of $\triangleleft$. We now use the duoidal structure for the object it was built to study.

9.1 Double comonoids

A duoidal category has *double comonoids*: objects carrying compatible comonoid structures for both tensors. For $(\mathbf{Cont}, \triangleleft, \otimes)$ the $\triangleleft$ -comonoids are exactly the directed containers — the small categories (Chapter [directed containers]; the identification is Ahman–Uustalu [9], and in $\triangleleft$ -comonoid form the $\mathcal{C} = 1$ case of Dorta–Jarvis–Niu [10, Thm. 4.3]) — so a double comonoid is

a small category with an extra, $\otimes$ -compatible comultiplication. Which small categories admit one?

The natural guess, from Spivak’s eq. (33) [Cited], is “the degenerate ones”: $c = y^A$ (one object) or $c = Ay$ (discrete), because those are exactly where the comparitor $o_{c,c}$ is invertible. The guess is wrong — in both directions — and the true answer is sharper and prettier.

Theorem 9.2 (Double comonoids are the sets of commutative monoids). *A directed container c is a double comonoid in the normal duoidal category $(\mathbf{Cont}, \triangleleft, \otimes)$ if and only if its underlying category is a coproduct of one-object categories on commutative monoids:*

$$c = \sum_I y^{M_I}, \quad \text{each } M_I \text{ a commutative monoid.}$$

When it exists the $\otimes$ -comonoid structure is unique: the counit is the family of identities, the comultiplication is the diagonal $I \mapsto (I, I)$ on objects and the monoid multiplication μ_I on each fibre. [MacBeth; the ingredients Cited as below]

Proof (the shape of it). By the “comonoids in comonoids” principle [8], a double comonoid is a $\otimes$ -comonoid object in the category of $\triangleleft$ -comonoids, and $\otimes$ lifts to that category (which is $\mathbf{Cat}^\sharp$, categories and cofunctors) as the *product of categories* [3, Prop. 8.79]. So the data are a cofunctor counit $c \rightarrow \mathbb{K}$ and a cofunctor comultiplication $c \rightarrow c \times c$, satisfying the comonoid laws. Three steps.

Step 1 (everything is an endomorphism). The counit’s lifting property forces the $\otimes$ -counit to select identities. The comultiplication’s object map is the diagonal (by the counit law), and the cofunctor codomain axiom applied to the diagonal forces $\text{cod}(g) = I$ for every arrow g out of every object I (take the second component to be an identity). A category all of whose arrows are endomorphisms is a coproduct of one-object categories — a *set of monoids* $c = \sum_I y^{M_I}$.

Step 2 (Eckmann–Hilton). On the fibre M_I the multiplication is a map $m_I: M_I \times M_I \rightarrow M_I$, and the cofunctor composition axiom says precisely that m_I is a homomorphism from the *product* monoid $(M_I \times M_I, \cdot \times \cdot)$ to $(M_I, \cdot)$, sharing the unit id_I . Two operations on a set with a common unit, one a homomorphism for the other, coincide and are commutative (Eckmann–Hilton): $m_I = \mu_I$ and M_I is commutative.

Step 3 (converse). Conversely, for a set of commutative monoids, define the counit by identities and the comultiplication by the diagonal and the fibre multiplications. The composition axiom is the equation $(g_1 h_1)(g_2 h_2) = (g_1 g_2)(h_1 h_2)$, which holds *because* each M_I is commutative; the comonoid laws are associativity and unitality of μ_I ; everything is per-fibre, so there is no cross-object obstruction. $\square$

Corollary 9.3 (The guess, corrected). *The degenerate polynomials are strict special cases, and the guess fails both containments:*

- y^A (one object, monoid A) is a double comonoid iff A is commutative — so y^{S_3} is not one, though the guess admits it;
- multi-object examples such as $2y^2 = \{\mathbb{Z}/2, \mathbb{Z}/2\}$ are double comonoids, though the guess excludes them.

Thus $\{\text{degenerate}\} \subsetneq \{\text{sets of commutative monoids}\} = \{\text{double comonoids}\}$. [MacBeth]

Remark 9.4 (What is cited and what is ours). Spivak owns the moving parts: the comparitor [4, eq. (32)], its invertibility locus eq. (33) — which is the degenerate diagonal — and the classification of $\otimes$ -comonoids as “sets of monoids” [4]. What is ours is (i) the descent analysis — the observation that a double comonoid needs the comultiplication to land in the order-independent image, which is available for *any* set of monoids, not only where o is invertible; (ii) the identification of the compatibility as a fibrewise Eckmann–Hilton square; and hence (iii) the corrected classification. So this is properly read as “eq. (33) on the diagonal, plus

the converse”, and the delta is the converse together with the commutativity it forces. The Eckmann–Hilton reduction is corroborated by an exhaustive machine check over a monoid library (§10); the proof itself rests on the two-line lemma, not the check.

9.2 The wider landscape

Beyond the comparitor and the double comonoids, the four structures relate in many ways — distributive laws, the $\otimes/\times$ comparison, the semicartesian $\triangleright_S$ family — and Jules Hedges has remarked informally that there are of the order of ten interesting relationships among them. We do not attempt a census of those here: the ones we can state at the grade this book demands are the interchange ζ (Theorem 9.1), the comparitor and its universal property (Theorem C), the closures (§8), and the double-comonoid classification (Theorem 9.2). A complete and honestly graded enumeration — matching each folklore relationship to a proof or a citation — is worthwhile future work, and we flag it as such rather than manufacture a list.

10 Provenance, machine-checking, and what is open

Cited, not ours

- Day convolution on **Poly**; the formula $p \odot q \cong \sum y^{p[s]*q[t]}$; distribution over coproducts; and the identifications $\times = \text{Day}(+, \emptyset)$, $\otimes = \text{Day}(\times, 1)$ — Niu–Spivak [3, Prop. 3.79, eqs. (3.80)–(3.81)]; Spivak [4].
- $A \vee_S B = A + (A \times S \times B) + B$ and the induced $\triangleright_S$ on **Poly** — Spivak [4, eqs. (9),(12)], who credits R. Garner; the $S = 1$ case is Niu–Spivak [3, Ex. 3.82]. (Monoidality is *asserted* there; the coherence proof is Proposition 6.2.)
- The comparitor $p \otimes q \rightarrow p \triangleleft q$, its lax monoidality, and its invertibility for representable q — Niu–Spivak [3, Ex. 6.84, Ex. 6.85, Prop. 6.87]; Spivak [4, eqs. (32),(33)]. The duoidal interchanger it is usually derived from — Spivak–Srinivasan [5].
- $\triangleleft$ is not cocontinuous in the right variable — Niu–Spivak [3, Ex. 6.56].
- The internal hom for $\otimes$ (Definition 8.1, Theorem 8.2) — Niu–Spivak [3, Ex. 4.78, eq. (4.79)]; Spivak [4, eq. (44)]; Spivak–Srinivasan [6, Prop. 2.8]. The three named convolutional closures — Spivak [4, eqs. (38)–(40)].
- The right coclosure of $\triangleleft$ (Theorem 8.6) — Spivak [4, eqs. (68)–(69)]; Niu–Spivak [3, Prop. 6.57], due to J. Meyers.
- **Cont** cartesian closed, not locally cartesian closed (Theorem 8.8) — Altenkirch–Levy–Staton [7]; the CCC exponential is Niu–Spivak [3, Thm. 5.31].
- That a $\triangleleft$ -comonoid is a small category (equivalently, a directed container is a polynomial comonad), with cofunctors for morphisms — Ahman–Uustalu [9]; in $\triangleleft$ -comonoid form the $\mathcal{C} = 1$ case of Dorta–Jarvis–Niu [10, Thm. 4.3]. We claim no part of this identification as new: only its Lean formalisation (`Comonoid.lean`, `ComonoidConverse.lean`, both round trips) is ours, and we believe it the first machine-checked proof of the equivalence.
- **(Cont, $\otimes$, $\triangleleft$)** normal duoidal (Theorem 9.1) — Spivak–Srinivasan [5, eq. (29)]. That $\otimes$ lifts to $\text{Cat}^\sharp$ as the product of categories — Niu–Spivak [3, Prop. 8.79]. Comonoids-in-comonoids — Aguiar–Mahajan [8].
- The comparitor’s invertibility locus eq. (33) and the classification of $\otimes$ -comonoids as “sets of monoids” — Spivak [4].
- Full faithfulness of $\llbracket - \rrbracket$ — Abbott–Altenkirch–Ghani [1].
- **Cont** $\cong$ **Fam(Set^{op})** — folklore in the polynomial-functor literature; Lemma 2.4 records an elementary proof and claims no originality.³

³Locators are deferred to a citation pass; we prefer no citation to a citation we have not personally verified.

- The identification of the two non-Day tensors of Dorta–Jarvis–Niu with the Dialectica products of Gödel’s functional interpretation has been *relocated* to the companion held file `papers/ductica-tensors-deferred.tex` (deferred to a later **Cont**(*C*) chapter); its citations and provenance travel with it.

New here

- **Theorem 4.7** — the classification. The sources state only the existence direction (“for any monoidal structure on **Set** there *is* one on **Poly**”). Theorem A supplies the converse and identifies the two conditions (D1), (D2) that cut out the essential image.
- **Lemma 4.6** — representability of the unit is a *consequence* of (D1)+(D2), so “convolutional” is a condition on the tensor alone.
- **Theorems 5.5, 5.6, 5.9** — the product is the unique pointwise monoidal structure on **Cont** (no Day hypothesis), with an effective test.
- **Lemma 5.4** — rigidity of the coproduct on **Set**; surely folklore, but we have not found it stated, and it is what makes (2) $\Rightarrow$ (3) of Theorem B go through.
- **Corollary 6.6** — a proper class of convolutional structures, all with terminal unit, exactly one cartesian: the family is unit-blind.
- **Theorem 7.1** — the comparitor is the counit of a coreflection, $p \otimes (-) = \text{Lan}_J((p \triangleleft -) \circ J)$. In the literature the comparitor is always *derived* from the duoidal interchanger; Theorem C says what it *is*, and the standard facts about it become corollaries.
- **Proposition 7.5** — the invertibility locus is *exactly* linear-or-representable; the sources give the sufficient direction only.
- **Proposition 7.6** — the comparitor is not injective either; the vacuous-shape phenomenon.
- The explicit container-level associator and unitors for $\triangleleft$, the flattening proof of the pentagon, and the flat-tuple argument on positions (§3.3).
- **Theorem 9.2** — the double comonoids of $(\mathbf{Cont}, \triangleleft, \otimes)$ are exactly the sets of commutative monoids, by a fibrewise Eckmann–Hilton collapse (Remark 9.4 scopes the delta against Spivak’s eq. (33): the converse direction and the commutativity it forces).
- **Remark 8.4** — the single uniform closure formula $[p, q]_\star = \prod_I q \triangleleft (p[I] \star y)$ specialising to Spivak’s three, graded [computed].
- The **Lean certification** of the Dirichlet closure (Remark 8.3); the formula is cited, the machine-checked closed monoidal category is, we believe, the first of its kind.
- The identification of DJN’s uninterpreted tensors $\ltimes, \rtimes$ with de Paiva’s Dialectica tensor and its directed variant — answering the open question of [10, §6] — has been *relocated* to the companion held file `papers/ductica-tensors-deferred.tex`, together with the closed forms, the non-convolutional placement, and the novelty sweep. It is deferred from the core programme, not withdrawn.

The closest neighbour

The nearest work in the literature is Dorta–Jarvis–Niu [10], and it is worth saying precisely where they meet us and where they do not. They study monoidal structures on the *generalized* polynomial categories $\Sigma\Pi\mathcal{C}$ — the free coproduct completion of the free product completion of a monoidal category $\mathcal{C}$, with objects the formal sums $\sum_i \prod_a c_{i,a}$ whose entries $c_{i,a}$ live in $\mathcal{C}$. Taking $\mathcal{C}$ terminal recovers exactly our setting: $\Sigma\Pi 1 \simeq \mathbf{Cont}$. They transport both the Day tensor $\otimes$ and the substitution tensor $\triangleleft$ up to $\Sigma\Pi\mathcal{C}$, and their Theorem 4.3 identifies the $\triangleleft$ -comonoids with enriched categories and enriched cofunctors — whose $\mathcal{C} = 1$ case is the Ahman–Uustalu statement, recalled above, that a $\triangleleft$ -comonoid is a small category. That theorem, not this chapter, is the source for the fact that a directed container is a $\triangleleft$ -comonoid; §3.3 and §9 use it only as standing background.

Their generalisation runs along an axis *orthogonal* to ours, which is why the three main theorems survive intact. Dorta–Jarvis–Niu vary the *predicate* axis — enriching each position and direction by an object of an arbitrary monoidal $\mathcal{C}$, so that $\Sigma\Pi 2$ is de Paiva’s $\text{Dial}(\mathbf{Set})$, and $\mathcal{C}$ metric gives weighted categories — and they *construct* the Day convolution (their Prop. 3.1) without classifying it: there is no equivalence, no uniqueness statement, and no universal characterisation of the comparator anywhere in their paper. We instead fix $\mathcal{C} = 1$ and vary the *base* axis — the monoidal structure on $\mathbf{Set}$ — and fill exactly that gap. Theorem 4.7 upgrades the Day construction from one direction to an equivalence $\{\text{monoidal structures on } \mathbf{Set}\} \simeq \{\text{convolutional structures on } \mathbf{Cont}\}$; Theorem 5.6 proves the product is the unique pointwise structure; Theorem 7.1 identifies the comparator as a coreflection counit. None of the three has a counterpart in [10]. Finally, their §6 records two further monoidal structures on $\Sigma\Pi \mathcal{C}$ (written $\ltimes, \rtimes$) that they cannot yet interpret — a concrete second instance of the “why exactly these?” question. On the $\mathbf{Set}$ side Theorem 4.7 answers it; the answer on *their* side — $\ltimes$ as de Paiva’s Dialectica tensor extended to all of $\mathbf{Poly}$, and $\rtimes$ as its directed variant — is developed in the companion held file `papers/ductectica-tensors-deferred.tex`, deferred to a later $\mathbf{Cont}(\mathcal{C})$ chapter.

Machine-checked

All four monoidal structures are formalised in Lean 4 as `MonoidalCategory` instances: `contCoproductMonoidal` and `contProductMonoidal` (`FourMonoidal.lean`), `contDirichletMonoidal` (`Dirichlet.lean`), `ContMonoidal` (`Monoidal.lean`); together with the extension laws $\llbracket p \triangleleft q \rrbracket = \llbracket p \rrbracket \circ \llbracket q \rrbracket$, $\llbracket p \times q \rrbracket \cong \llbracket p \rrbracket \times \llbracket q \rrbracket$, and the comparator `Container.dirToSeq` with its non-surjectivity. The sequential pentagon and triangle (Theorem 3.5) close by `rfl`. The **Dirichlet closure** (Theorem 8.2) is formalised in full in `DirichletClosed.lean`: the internal hom, the currying bijection with both round trips, and naturality in each variable, all `rfl` and axiom-free.

The finite-container computations behind §6 — the $\vee_S$ pentagon and triangle, the κ test, the comparator trichotomy, and the negative control of §6.1 — are Python, not Lean; bijections are constructed and compared elementwise rather than counted. The uniform closure formula (Remark 8.4) is checked on 5197 finite triples with three negative controls; the Eckmann–Hilton reduction behind Theorem 9.2 is corroborated by an exhaustive search over a monoid library (commutative $\Rightarrow$ a unique $m = \mu$; S_3, T_2 admit none). **Not yet machine-checked, and flagged as such:** Theorems A, B^+ , C, and 9.2 themselves; the $\vee_S$ coherence proof; the uniform closure adjunction in the generality of Remark 8.4.

Open

Question 10.1 (Naturality-free Theorem B). Theorem 5.5 assumes the pointwise isomorphism is natural in p and q . Suppose only that for each p, q there *exists* an isomorphism $\llbracket p \odot_\star q \rrbracket \cong \llbracket p \rrbracket \times \llbracket q \rrbracket$ natural in X alone. Then $A \star B \cong A + B$ for all A, B as bare bijections, and Proposition 5.2 still gives $I = \emptyset$, so κ exists — but we cannot show κ is bijective without naturality. *Is there a monoidal structure $(\star, \emptyset)$ on $\mathbf{Set}$ with $A \star B \cong A + B$ for all A, B but $\star \not\cong +$ as a bifunctor?* We believe not, and have no proof. Nothing above depends on the answer. [Open]

Question 10.2 (Symmetry of $\vee_S$). Is $(\vee_S, \emptyset)$ symmetric? The candidate braiding is evident; the hexagon is unverified (Remark 6.3), and the negative control of §6.1 is a specific reason for caution, since a braiding must re-match separators. [Open]

Question 10.3 (Closed convolutional structures). Is a convolutional tensor $\odot_\star$ closed *only if* $R\star(-)$ is polynomial? Remark 8.4 gives the sufficiency; the necessity is unargued (Remark 8.5), and with it Theorem A would upgrade to an equivalence between $\star$ with $R\star(-)$ polynomial and the closed convolutional structures on $\mathbf{Cont}$. [Open [speculative]]

Question 10.4 (A complete duoidal atlas). Enumerate, at book grade, the interaction laws among the four structures — matching each of the folklore relationships (§9) to a proof or a citation, and settling which are open. [Open]

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